Rhea Madhogarhia
Final Paper | Phil of AI, Callard & Bridges
Due: Dec 15th (extension)

Limits of Matter and Manner

The emergence of "mechanistic interpretability", the study of how neural networks compute their outputs, rests on the idea that, at the moment, the "black box" description of generative AI systems isn't sufficient. Essentially, however complex artificial intelligence models are, reverse engineering their outputs can reduce the opacity of generative machine learning models and show a mechanical, identifiable process of understanding or thinking that takes unseen inputs and transforms them into intelligible, and naturalistic-seeming outputs. GPT models have been passing the Turing Test, and we speak about AI as if it is seeing, painting, drawing, thinking, learning, and communicating; our communication with chatbots has even led us to see the production of these outputs as a type of companionship, arguably an anthropomorphism of artificial intelligence. Still, more than three-quarters of US adults are either unsure of the benefit or sure that AI may be harmful. So, why do we tend to anthropomorphize and rationalize AI systems? Is this tendency justifiable, or is it a blinding human tendency that traps us into believing that humans and AI are intellectually matched?

In this paper, I will argue that the architectural complexity (the construction) of Generative AI forces us to understand these systems as intentional out of pragmatic necessity – to treat them as unintentional would be to call their outputs irrational or untrustworthy. But, this pragmatic necessity masks the failure of Generative AI to satisfy a construction-level characterization required for understanding, intentionality, and rationality. The tension between the complexity and opacity of AI models and the pragmatic necessity we have to understand and engage with them is what traps us into the illusory anthropomorphization of AI as rational and intentional interlocutors.

In his essay, “Intentional Systems”, Daniel Dennett defines Intentionality as any system "whose behavior can be (at least sometimes) explained and predicted by relying on ascriptions to the system of beliefs and desires". He claims that intentionality is “observer dependent” – that intentionality is defined only within that perspective of the inquiring party. He also assumes intentionality as a “feature of linguistic entities”, which allows me to include Large Language Models in the discussion of intentionality. And, finally, Dennett assumes that the system in question is behaving rationally given its goals. Essentially, to seriously inquire about a system’s Intentionality, until you end your investigation, you should assume the system’s rationality says something about how to predict its behavior. I wonder if this is because Dennett contends that rationality is something we either have not yet been able to define or is less philosophically problematic than Intentionality. In other words, Intentionality is some higher barrier to the human condition that a system must surpass to be comparable to the humankind.

Dennett also situates the Intentional Stance in opposition to the Physical and Design Stances as strategies for classifying and explaining a system’s behavior. The proposed distinction between these stances is the mechanism of prediction. Prediction is enabled by the laws of nature by the Physical Stance, the functional design of the system by the Design Stance, and the assumption of rationality by Dennett’s Intentional Stance. Thus, only a rational system, where rationality is explicitly *not* definable through the Physical or Design Stances, can be intentional. So, under Dennett’s definitions, to determine if AI is Intentional, we must first believe that an AI’s output should be assessed on the basis of rationality.

Dennett does not undertake defining why doing the optimal thing given a specific goal is a sufficient description of rationality in this essay, a smart move considering the difficulty of that inquiry for mankind, the rational kind. The behavior of the system is thus seen as something too complex, or too massive of a concept for the Physical and Design Stances to define: brains and computers are both systems in which there are too many physical variables for us to keep track of, thus failing to be explained by the Physical Stance, and in which a brain or computer’s internal states have too intricate of a representation to be described by the Design Stance. Whether or not the Physical or Design Stance is intelligible is unimportant because they both fail to be pragmatic stances. Humans don't have the computational power to predict a complex system's behavior physically – this would be "a pointless and herculean labor" according to Dennett. Similarly, the internal operations of the human biological system or Large Language Model are so layered and multimodally impacted that predicting behavior becomes cognitively impossible through Design Stance inspection.

Importantly, these failures of the Design and Physical Stances are unfortunate to Dennett because they are greatly pragmatic stances for less complex systems; they are only limited by *our* own computational and biological limitations. But, it is exactly those limitations that Dennett believes make the Intentional Stance the pragmatic, or at least justifiable, approach to studying behavior. How can we claim to have a theory of behavioral prediction that our own behavior (our limitations) prevents us from using? Dennett believes we should have no such theory. Thus, the value of the Intentional Stance is an acceptance of our limited capacities, using our ability to exhibit rationality through our behavior to produce a reasonable theory of behavioral prediction.

To examine a system through the Intentional Stance is to only use our human-granted rationality without having to discuss its substance or the existence of consciousness.

Dennett makes an arguably compelling argument for the viability of the Intentional Stance and, for the reasons I discussed above, proves the Intentional Stance to be a pragmatic necessity for understanding complex systems. Notably, the Intentional Stance does not metaphysically grant AI a soul or a human likeness, but offers a strategy to evaluate predictive behavior. The Intentional Stance seems to be quite similar to how we already evaluate AI. We don't say "ChatGPT is acting as if it understood my question" when it gives us satisfying answers; we say "ChatGPT understood my question". Thus, the Intentional Stance seems to be a natural mode of investigation for us. This is good for us, considering the Physical and Design Stances' practical difficulties. But, is the pragmatic utility or practical success of the Intentional Stance enough to let it say something about the behavior of understanding?

Jason Bridges argues that genuine understanding requires an “activity-level” and “construction-level” characterization of the system in his paper, “Meaning & Understanding”. This description responds to an intuitive definition of understanding: the Guidance Conception says that understanding is the process of having something “come before the mind” to guide one’s actions. The major flaw in the Guidance Conception, for Bridges, is that the directions of a guide must be understood by the system following them. So, we cannot use a guide to define understanding if understanding is part of the process needed to follow a guide. Ironically, this seems similar to the structure of Dennett’s defense of the intentional Stance: we cannot use the Physical or Design stance to define understanding if we are unable to understand how to pragmatically use these stances. form of correctness, helpfulness, or precision. This characterization is context-dependent, and though necessary, only describes “accordance”, or what many of my computer scientist friends would call "spurious correlations”. Thus, the construction-level characterization provides a context-independent description of the understanding system in question. The description of “apparatus” or architecture of an internal state of a system (construction-level) must correspond to the observed understanding at the activity-level under Bridges' requirements for an understanding Mechanism.

When a Large Language Model (LLM) produces an output, we ought to say, by the Intentional Stance, that the LLM decided, or reasoned, to answer our question. In other words, the LLM was guided towards an answer by its own rationality. For Bridges, this is an activity-level characterization. By Dennett’s design, the activity-level characterization contains a projection of our own reality onto a Mechanism’s outputs, but Bridges argues a construction-level characterization must be provided. The field of mechanistic interpretability seems to provide this: examinations of the weights, biases, and layered architecture of the neural networks that underlie LLMs. But this construction-level characterization is unsatisfactory because it arguably remains context-dependent (although that is not the goal of mechanistic interpretability researchers). If mechanistic interpretability is only able to use the context-rich activity-level characterizations to support construction-level characterization, then we do not isolate the apparatus of a model from its context. If this field of study continues on to successfully find inherent conceptual meaning in the states of AI models, generalizable across different machine learning architectures, then mechanistic interpretability will be the true study of computational understanding.

To make this conflict tangible, consider the activity-level characterization of an LLM's output: that it "gave a reasonable or helpful response" to a question. The supposed construction-level accounts are the states of a neural network: matrices of numbers and coefficients (weights) of variables. There is no context-independent mapping of “understanding” or “helpfulness” to a matrix of numbers. Recall that Dennett defined Intentionality as a feature of linguistic entities, whereas the LLM's actual mechanism is purely numerical. Thus, the LLM satisfies the conditions for Intentionality as a pragmatic simulation. The construction level remains mathematically/numerically abstract rather than conceptually grounded, so the system is fully secluded to activity-level characterizations. The disconnect here is that the Mechanism produces the *outputs* of understanding without satisfying the conditions for a complete *process* of understanding required by Bridges.

I am unsure of whether or not humans pass Bridge’s test, but it seems silly to say that we are projecting a higher form of understanding to understand our own... understanding. Still, this seems to be the project of the biological sciences and scientific inquiry in general. In this essay, I have tried to argue why AI does not satisfy Bridges requirements for understanding. So far, we have yet to reach satisfying construction-level characterizations of AI systems, leaving us with an activity-level characterization that projects rationality onto these systems — the aforementioned anthropomorphization. The complexity of AI systems, like ChatGPT, forces us into taking the Intentional stance. But, the opacity that comes with architectural complexity also prevents us from identifying a construction-level characterization. The more complex AI models get, the more we get backed into a corner granting AI a "pinched and artificial" rationality. And we are thus pushed even further from crafting a construction-level understanding. Our anthropomorphism of AI is explainable by the limitations put on our brains and bodies by cognitive capacity, a testament towards *only* humankind's Intentional and understanding capacities.

- Rhea

Thank you for an awesome quarter!

Extra:

I put my essay into Gemini, asking it to tell me what it thinks my argument/conclusion is (a quick 2 am sanity check), and here is how it summarized (but also confirming here that I did not use Gen AI to write any of the above!). You can tell that it took "conclusion" literally since it seems to directly summarize my last 4-5 sentences.

- "The human mind is backed into an illusory corner: the very complexity that makes the AI seem intelligent is the same complexity that prevents us from verifying its intelligence. Our anthropomorphism of AI is ultimately a testament to the success of LLM design in perfectly mimicking the output of reason, and a testament to our own cognitive capacities, which demand a shortcut when faced with overwhelming complexity.